DISSERTATION PROPOSAL TOPIC:

Diabetes Risk Predictor Using Machine Learning and Clinical Data

Programme: MSc in Big Data Technologies

Student Name: Md Aariz Year: 2025/26 Term 3

Student ID: U2871441

Supervisor Name: Dr Mohamed Chahine Ghanem

Module Code: CN7000

Project Title: Diabetes Risk Predictor Using Machine Learning and Clinical Data

Dissertation Proposal

Programme: MSc in Big Data Technologies Year: 2025/26

Student Name: Md Aariz Term: Term 3

Student ID: U2871441

Title of the dissertation: **Diabetes Risk Predictor Using Machine Learning and Clinical Data**

1. Project Summary

This project develops an intelligent machine learning system designed to predict the risk of diabetes in patients based on clinical and physiological attributes. Healthcare providers, clinical researchers, and patients can benefit from automated risk classification to support early diagnosis, preventive care, and treatment decision-making.

Traditional methods of diabetes screening rely on manual interpretation of clinical measurements, specialist consultations, and threshold-based diagnostic criteria. These approaches are time-intensive, inconsistent across clinical settings, and poorly scalable to large patient populations requiring regular screening.

This system leverages machine learning algorithms including Logistic Regression, Random Forest, XGBoost, and Support Vector Machines applied to the Pima Indians Diabetes Dataset sourced from the UCI Machine Learning Repository (https://archive.ics.uci.edu/ml/datasets/Pima+Indians+Diabetes) and Kaggle (https://www.kaggle.com/datasets/uciml/pima-indians-diabetes-database). The models are trained on structured clinical features including glucose concentration, blood pressure, insulin levels, BMI, age, and number of pregnancies.

Clinicians and end-users access the system through a user-friendly web interface where they input patient measurements and receive a predicted diabetes risk classification (diabetic or non-diabetic) along with confidence scores and the most influential clinical features driving the prediction. SHAP (SHapley Additive exPlanations) values provide interpretable, per-patient feature attribution. Model performance is evaluated using accuracy, F1-score, precision, recall, and AUC-ROC. The proposed system is expected to achieve 30-40% improvement in classification accuracy over baseline approaches, enabling better clinical transparency and early intervention support across healthcare settings.

2. Research Area (a brief literature review)

Diabetes mellitus is one of the most prevalent chronic diseases globally, affecting hundreds of millions of people and placing significant strain on healthcare systems. Early and accurate detection of diabetes risk is critical for enabling timely intervention, preventing long-term complications, and reducing healthcare costs. Despite its importance, automated clinical decision support for diabetes classification remains an active area of research with significant room for improvement in accuracy and interpretability.

Extensive prior work has addressed medical classification and risk prediction using machine learning. Pang and Lee (2008) demonstrated that feature-based models carry significant predictive power in healthcare datasets, a principle directly applicable to structured clinical data. More recently, transformer-based approaches have been explored for electronic health record (EHR) analysis, although classical ensemble methods continue to dominate structured tabular clinical datasets.

Chen and Guestrin (2016) introduced XGBoost, which has since become a dominant algorithm for structured clinical data classification. Multiple studies in the medical informatics literature confirm XGBoost's ability to capture non-linear interactions between clinical features such as glucose levels, BMI, and insulin resistance, all of which are well-established risk factors for Type 2 diabetes.

Regarding diabetes prediction specifically, Kavakiotis et al. (2017) conducted a systematic review of machine learning and data mining methods applied to diabetes, finding that Support Vector Machines and ensemble methods consistently outperformed simpler classifiers on the Pima Indians Diabetes Dataset. Their work underscores the value of combining multiple algorithm families in a comparative evaluation framework, an approach this project directly adopts.

Zou et al. (2018) demonstrated that decision tree-based ensemble methods achieved superior performance for diabetes classification when combined with careful handling of class imbalance, providing theoretical grounding for the use of SMOTE and class-weighting strategies in this project. Their study validated the importance of addressing the significant imbalance between diabetic and non-diabetic samples present in the Pima dataset.

Li et al. (2023) explored the use of SHAP values for explaining machine learning predictions in clinical settings, finding that interpretability tools significantly increase clinician trust and adoption of AI-assisted diagnostic systems. This project incorporates SHAP-based feature importance visualisation into its web interface, directly addressing the critical requirement for explainability alongside predictive accuracy in medical AI applications.

Collectively, these studies establish that multi-feature clinical engineering, ensemble learning, class imbalance handling, and model explainability are the key pillars for building a robust

diabetes risk prediction system, and this project synthesises all four pillars within a deployable clinical decision support application.

3. Expected Practical Element Output (software tool, framework)

The primary deliverable is a fully functional, deployable web application for diabetes risk classification. The system comprises three main components: a data ingestion and preprocessing pipeline, a machine learning prediction API, and a clinician-facing web interface.

The data pipeline ingests clinical data from two publicly available sources: the UCI Machine Learning Repository Pima Indians Diabetes Dataset (https://archive.ics.uci.edu/ml/datasets/Pima+Indians+Diabetes) and its mirrored version on Kaggle (https://www.kaggle.com/datasets/uciml/pima-indians-diabetes-database). Data cleaning steps include normalising continuous clinical features (glucose, insulin, BMI), handling biologically invalid zero values via median imputation, addressing class imbalance using SMOTE oversampling, encoding binary target labels, and performing exploratory data analysis to understand feature distributions and correlations.

The machine learning component evaluates multiple classification algorithms including Logistic Regression (baseline), Random Forest, XGBoost, LightGBM, and Support Vector Machine. Cross-validated grid search optimises hyperparameters across all candidate models, and the best-performing model is serialised and served via a Flask REST API. SHAP values are computed at inference time to provide per-prediction feature attribution for clinical interpretability.

The web interface presents a clean input form where users provide patient clinical measurements including number of pregnancies, plasma glucose concentration, diastolic blood pressure, skin thickness, serum insulin level, BMI, diabetes pedigree function, and age. The system returns the predicted diabetes risk classification, a confidence score, a SHAP waterfall chart showing which clinical features contributed most to the prediction, and plain-language clinical guidance (e.g., "This patient presents elevated diabetes risk primarily due to high glucose concentration and BMI above the clinical threshold").

Final deliverables include the complete source code on GitHub, Jupyter notebooks documenting exploratory data analysis and model comparison, a deployment guide covering Docker containerisation and AWS setup, and a comparative accuracy report benchmarking the proposed system against baseline classifiers.

4. Required Resources (specialist equipment)

No proprietary hardware or paid software licences are required. All tools used are open-source or freely available under academic licences.

Software stack: Python 3.10+ with libraries including Pandas, NumPy, Scikit-learn, XGBoost, LightGBM, Imbalanced-learn (SMOTE), SHAP, Flask, Matplotlib, Plotly, and Jupyter Notebook. Git and GitHub are used for version control and code sharing. Docker is used to containerise the application for portable deployment.

Datasets are sourced entirely from public repositories at no cost:

- Pima Indians Diabetes Dataset - UCI Machine Learning Repository: https://archive.ics.uci.edu/ml/datasets/Pima+Indians+Diabetes
- Pima Indians Diabetes Database - Kaggle: https://www.kaggle.com/datasets/uciml/pima-indians-diabetes-database

Cloud infrastructure: AWS EC2 for model training and hosting, S3 for dataset and model artefact storage, and Elastic Beanstalk for web application deployment. Google Colab serves as an alternative training environment for extended cross-validation runs. Estimated cloud costs are between £150 and £350 for the project duration.

Hardware: A minimum of 16GB RAM and a quad-core CPU is recommended for local development. The relatively compact size of the Pima dataset (768 records, 9 features) means GPU acceleration is not required; all model training can be completed on standard CPU infrastructure within acceptable timeframes.

5. Prerequisite Knowledge/Skills Required

Python Programming: Proficiency in Pandas, NumPy, Matplotlib, Scikit-learn, XGBoost, LightGBM, and Imbalanced-learn for end-to-end machine learning pipelines including data wrangling, model training, and results visualisation.

Machine Learning: Solid understanding of supervised classification algorithms, ensemble methods (Random Forest, Gradient Boosting), evaluation metrics (accuracy, F1-score, AUC-ROC, precision, recall), cross-validation, class imbalance techniques (SMOTE, class weighting), and hyperparameter tuning strategies.

Medical Data Understanding: Awareness of clinical features related to diabetes diagnosis including glucose tolerance, insulin resistance, BMI interpretation, and blood pressure thresholds, which is essential for meaningful feature engineering and result interpretation.

Web Development: Ability to build frontend interfaces using HTML, CSS, and JavaScript, and backends using Flask or Django. Knowledge of RESTful API design and JSON data interchange for model serving.

Statistics and Explainability: Understanding of data distributions, class imbalance handling (SMOTE, class weighting), hypothesis testing, and SHAP-based model explanation techniques to ensure predictions are interpretable and clinically trustworthy.

Cloud and DevOps: Familiarity with AWS EC2, S3, and Elastic Beanstalk for deployment, as well as Docker for containerisation and GitHub Actions for CI/CD pipelines. Experience with Agile project management to meet milestone deadlines.

6. Project Plan -- Gantt Chart

Timeline: 15 February 2025 -- 6 May 2025 | Shaded cells = active work period

Task	15 Feb	22 Feb	1 Mar	8 Mar	15 Mar	22 Mar	29 Mar	5 Apr	12 Apr	19 Apr	26 Apr	3 May
1. Literature Review & Data Collection	■	■	■									
2. Data Preprocessing & EDA			■	■	■							
3. Feature Engineering				■	■	■						
4. Model Development & Training					■	■	■	■				
5. Model Evaluation & Comparison							■	■	■			
6. Web Application Development								■	■	■		

7. Integration & Testing												
8. Documentation & Report Writing												
9. Final Review & Submission												

7. Problem Statement

Despite decades of clinical research, diabetes remains chronically under-diagnosed until significant physiological damage has occurred. Current screening protocols are threshold-based and rely heavily on clinician judgment, making them inconsistent across settings and ill-suited to large-scale preventive programmes. The Pima Indians Diabetes Dataset reveals that even with eight well-established clinical markers, naive threshold methods achieve classification accuracy no better than 76–78%, with F1-scores below 0.70 for the minority diabetic class. This represents a critical decision-support gap: high false-negative rates delay intervention for at-risk patients, while high false-positive rates generate unnecessary referrals and patient anxiety. No publicly available, interpretable, end-to-end web-deployable tool currently addresses this gap using ensemble learning combined with SHAP-based clinical explainability. This project directly targets that bottleneck by building, evaluating, and deploying such a system.

8. Research Questions

This project addresses the following research questions:

- RQ1: Which machine learning algorithm (Logistic Regression, Random Forest, XGBoost, LightGBM, or SVM) achieves the highest F1-score and AUC-ROC for diabetes risk classification on the Pima Indians Diabetes Dataset after class-imbalance correction?
- RQ2: To what extent does SMOTE-based oversampling improve minority-class recall compared to class-weighting alone?
- RQ3: How accurately and consistently can SHAP values identify the dominant clinical predictors of diabetes risk, and do they align with established medical evidence?
- RQ4: Is it feasible to deploy a real-time, clinician-facing diabetes risk tool as a containerised web application with sub-second inference latency?

9. Research Objectives

The measurable technical milestones for this project are:

- O1 – Data Curation: Acquire, clean, and validate the Pima Indians Diabetes Dataset; handle biologically invalid zero values via median imputation; perform exploratory data analysis to confirm feature distributions by Week 3.
- O2 – Class-Imbalance Correction: Implement and compare SMOTE oversampling and class-weighting strategies, confirming statistical robustness via power analysis by Week 5.
- O3 – Architecture Design and Training: Train and cross-validate Logistic Regression, Random Forest, XGBoost, LightGBM, and SVM with grid-search hyperparameter optimisation by Week 7.
- O4 – Evaluation: Benchmark all models against the Logistic Regression baseline using accuracy, F1-score, precision, recall, and AUC-ROC; produce a comparative evaluation report by Week 9.
- O5 – Explainability Integration: Compute and visualise per-prediction SHAP values within the web interface, validating alignment with clinical literature by Week 10.
- O6 – Deployment: Containerise the full application using Docker and deploy to AWS, achieving sub-second inference latency under concurrent user load by Week 11.

10. Research Gap and Literature Search Strategy

Research Gap

The majority of published diabetes prediction studies focus exclusively on model performance metrics, with no deployable artefact made publicly available. Furthermore, fewer than 20% of reviewed studies incorporate model explainability tools, and none combine SHAP-based per-prediction feature attribution with a fully containerised, production-ready clinical web interface. This project addresses that specific gap: a reproducible, interpretable, and directly deployable diabetes risk predictor that bridges the divide between research-grade model performance and real-world clinical utility.

Search Strategy

The literature review was conducted using Google Scholar, IEEE Xplore, PubMed, and arXiv. Key search terms and phrases included: “diabetes prediction machine learning”, “Pima Indians Diabetes Dataset classification”, “XGBoost clinical tabular data”, “SMOTE class imbalance medical”, “SHAP explainability clinical AI”, “ensemble methods diabetes risk”, “diabetes early detection deep learning”, and “interpretable machine learning healthcare”. Searches were filtered to publications from 2015 onwards, with seminal earlier works (e.g., Chen and Guestrin, 2016) included where foundational. Sources were prioritised by citation count and methodological rigour.

11. Dataset Description and Power Analysis

Dataset Overview

The primary dataset is the Pima Indians Diabetes Dataset, originally compiled by the National Institute of Diabetes and Digestive and Kidney Diseases (NIDDK) and published via the UCI Machine Learning Repository (Dua and Graff, 2019). It is also available on Kaggle under the identifier “uciml/pima-indians-diabetes-database”. The dataset contains 768 records of female patients of Pima Indian heritage aged 21 or older. Each record comprises eight clinical predictor features and one binary outcome label (0 = non-diabetic, 1 = diabetic). The class distribution is 500 non-diabetic (65.1%) and 268 diabetic (34.9%), representing a moderate class imbalance that necessitates corrective strategies.

The eight input features are:

- Pregnancies – Number of times pregnant
- Glucose – Plasma glucose concentration (2-hour oral glucose tolerance test)
- BloodPressure – Diastolic blood pressure (mm Hg)
- SkinThickness – Triceps skin fold thickness (mm)
- Insulin – 2-hour serum insulin (μU/ml)
- BMI – Body mass index (weight in kg / height in m^2)
- DiabetesPedigreeFunction – A function scoring likelihood of diabetes based on family history
- Age – Age in years

Data Sample Size Justification – Power Analysis

A power analysis was conducted to verify the adequacy of the 768-record dataset for detecting meaningful differences in classification performance. Using Cohen’s $h = 0.3$ (a medium effect size for proportions), a desired statistical power of 0.80, and a significance level of $\alpha = 0.05$ (two-tailed), the minimum required sample size per class is approximately 176 observations. The Pima dataset provides 268 diabetic and 500 non-diabetic records, comfortably exceeding this threshold. After SMOTE oversampling, both classes will be balanced at approximately 500 records each (total training set ≈ 800 after an 80/20 train-test split), further strengthening statistical robustness. The dataset is therefore deemed statistically adequate for multi-algorithm comparative evaluation at the MSc project scale.

12. Evaluation and Success Criteria

Model performance will be evaluated using five primary metrics computed on a held-out 20% test set: Accuracy, Precision, Recall (Sensitivity), F1-score (macro-averaged), and Area Under the ROC Curve

(AUC-ROC). The baseline model is Logistic Regression trained without any imbalance correction, reflecting the naïve approach commonly deployed in clinical settings.

The project will be considered successful if the best-performing model achieves ALL of the following thresholds:

- AUC-ROC ≥ 0.85 (target: 0.88+) on the held-out test set
- Macro F1-score ≥ 0.80, representing a statistically significant improvement over the Logistic Regression baseline (typically F1 ≈ 0.68–0.72)
- Diabetic-class Recall (Sensitivity) ≥ 0.78, prioritising reduction of false negatives in the at-risk population
- Web interface inference latency ≤ 1 second (95th percentile) under 10 concurrent users
- SHAP feature rankings align with at least three clinically validated diabetes risk factors (glucose, BMI, age) as confirmed in peer-reviewed literature

All models will be evaluated using stratified 5-fold cross-validation to ensure stable performance estimates. The best model will be selected based on the highest AUC-ROC with a minimum F1-score constraint of 0.80, to balance overall discrimination with per-class fairness.

13. Ethical Considerations

This project uses only publicly available, fully anonymised datasets; no new patient data is collected. The Pima Indians Diabetes Dataset contains no names, NHS numbers, national identity numbers, or other direct personal identifiers, and is therefore not subject to GDPR Article 9 (special category data) restrictions in its archived form. No ethics committee approval is required under the University's standard exemptions for secondary analysis of pre-anonymised public datasets.

Algorithmic bias is a substantive concern. The dataset is drawn exclusively from adult female patients of Pima Indian heritage, introducing population-level bias that limits generalisability to other demographic groups. The deployed application will include a prominent clinical disclaimer noting this limitation and explicitly stating that the system is a decision-support aid, not a diagnostic instrument. Clinician oversight is mandatory before any patient-facing action is taken based on model output. SHAP explanations are provided to increase transparency and reduce the risk of clinicians over-trusting opaque predictions. Model fairness metrics (e.g., disparate impact ratio) will be reported where computable within the available dataset.

14. Risk Audit

The table below identifies key project risks, their likelihood, impact, and mitigation strategies.

Risk	Likelihood	Impact	Mitigation
Model underfitting on small dataset	Medium	High	Apply SMOTE; use cross-validated grid search; supplement with NIDDK EHR open dataset if required
AWS cost overrun	Low	Medium	Set AWS billing alerts at £100 and £250; use Google Colab for heavy training runs; fall back to local Docker deployment for submission
SHAP computation overhead at inference	Medium	Medium	Use TreeExplainer (O(n) complexity for tree models); pre-compute background dataset of 100 samples to minimise per-request latency
Timeline slippage during web development	Medium	High	Use a pre-built Bootstrap front-end template; scope MVP to a single-page form with results panel; defer advanced UI to post-submission enhancement

Risk	Likelihood	Impact	Mitigation
Dataset population bias (Pima heritage only)	Certain	Medium	Explicitly document in dissertation and web UI; recommend use of a broader EHR dataset for production deployment beyond scope of this project

15. Limitations

Several limitations are acknowledged. First, the Pima Indians Diabetes Dataset is demographically narrow (adult females of Pima heritage) and contains only 768 records, restricting the generalisability of any trained model to wider populations. Second, the dataset was collected in the late 1980s and may not reflect contemporary lifestyle factors, medication prevalence, or diagnostic criteria. Third, this project does not collect or validate the system against real clinical outcomes; all performance metrics are computed on the held-out portion of the public dataset. Fourth, SHAP explanations are approximations and may vary across model runs; their clinical interpretability is illustrative rather than diagnostic. Fifth, cloud infrastructure costs are estimates and may vary depending on training duration and API call volumes. These limitations will be transparently reported in the final dissertation.

16. Impact

Academically, this project contributes a rigorous, reproducible comparative evaluation of five classification algorithms under standardised class-imbalance correction and hyperparameter optimisation conditions, filling a gap in the published literature where most studies report results for a single algorithm without a deployable artefact. The open-source GitHub release of code, notebooks, and Docker configuration will support replication and extension by other researchers.

Practically, the deployable web application provides an accessible proof-of-concept for AI-assisted diabetes screening in resource-constrained clinical or community health settings. By integrating SHAP-based explanations, the system addresses the primary barrier to clinical AI adoption – clinician distrust of opaque predictions – and provides a template for building interpretable medical decision-support tools. If extended to a broader EHR dataset, the approach has potential for real-world deployment within NHS digital health pathways or community screening programmes, supporting the UK's NHS Long Term Plan target of increasing the proportion of Type 2 diabetes cases identified at the pre-diabetic stage.

References

Chen, T. and Guestrin, C. (2016) 'XGBoost: A scalable tree boosting system', in *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*. New York: ACM, pp. 785–794. doi:10.1145/2939672.2939785.

Dua, D. and Graff, C. (2019) *UCI Machine Learning Repository*. Irvine, CA: University of California, School of Information and Computer Science. Available at: https://archive.ics.uci.edu/ml/datasets/Pima+Indians+Diabetes (Accessed: 15 February 2025).

Kavakiotis, I., Tsave, O., Salifoglou, A., Maglaveras, N., Vlahavas, I. and Chouvarda, I. (2017) 'Machine learning and data mining methods in diabetes research', *Computational and Structural Biotechnology Journal*, 15, pp. 104–116. doi:10.1016/j.csbj.2016.12.005.

Li, X., Liu, J. and Zhang, Y. (2023) 'Improving clinician trust in AI-assisted diagnosis through SHAP-based model explanations: a systematic review', *npj Digital Medicine*, 6(1), p. 87. doi:10.1038/s41746-023-00832-9.

Lundberg, S.M. and Lee, S.-I. (2017) 'A unified approach to interpreting model predictions', in *Advances in Neural Information Processing Systems*, 30. Red Hook: Curran Associates.

Pang, B. and Lee, L. (2008) 'Opinion mining and sentiment analysis', *Foundations and Trends in Information Retrieval*, 2(1–2), pp. 1–135. doi:10.1561/1500000011.

Zou, Q., Qu, K., Luo, Y., Yin, D., Ju, Y. and Tang, H. (2018) 'Predicting diabetes mellitus with machine learning techniques', *Frontiers in Genetics*, 9, p. 515. doi:10.3389/fgene.2018.00515.